

 FIAT CHRYSLER AUTOMOBILES	RESISTANCE SPOT WELDING PROCESS LAUNCH PROCEDURES, VALIDATIONS AND CERTIFICATIONS FOR BODY-IN-WHITE OPERATIONS.	PS-A0215 Process Standard Page: 1/24 Date: 08-APR-2021
REGIONAL DOCUMENT		
Change level	Date	Description of change
-	22-JAN-2017	Initial release - ported over from ME BIW spec 389-08 Welding
A	01-SEP-2017	Updated section for transformer sizing requirements, weld schedule development and tip dressing/C-Factor process.
B	26-JUN-2018	Updated C-Factor window percentages, added requirements to provide weld cap trays, revised gun seating force, automated cap changer requirements. Added section 4.6 for weld reprocess procedures.
C	30-MAY-2019	Section 4.2.7 under "Notes for Table 4", Note #5 – Changed the required GMT from 1.50mm to 1.51mm
D	23-JUL-2019	Modified Table 5
E	03-JAN-2020	Clarified Tip Dresser requirements in Section 4.2.1
F	08-APR-2021	Section 1 – Removed supersede direction over other specifications and standards. Removed all welding activities outlined within the proceeding specification are to be reported as a separate line item on the line integrator quote. (Kaizen # 40204) Updated section 4.2.6 spare consumables verbiage Section 4.6 - Added deburring clarification for HS and AHS steels (Kaizen #27716) Section 6 - Added in weld expulsion quality concerns (PFJ 18175).
ANY PRINTED COPY IS TO BE DEEMED AS UNCHECKED; THEREFORE THE UPDATED COPY MUST BE CHECKED IN THE APPROPRIATE WEB SITE		
CONFIDENTIAL THIS DOCUMENT MUST NOT BE REPRODUCED OR CIRCULATED TO THIRD PARTIES WITHOUT PRIOR WRITTEN CONSENT BY THE RELEVANT FCA COMPANY. IN CASE OF DISPUTE, THE ONLY VALID REFERENCE IS THE ENGLISH EDITION		

RESISTANCE SPOT WELDING PROCESS LAUNCH PROCEDURES, VALIDATIONS AND CERTIFICATIONS FOR BODY-IN-WHITE OPERATIONS.

TABLE OF CONTENTS

1 GENERAL	3
1.1 Purpose	3
1.2 Coverage of this Standard (Applicability) and Limitations on Usage)	3
2 REFERENCES	3
3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS	4
4 PROCESS AND PROCESS CONTROL REQUIREMENTS	4
4.1 Condition of the part prior to processing	4
4.2 Resistance Spot Welding	4
4.2.1 RSW Design	4
4.2.2 RSW Documentation	8
4.2.3 RSW Build & Equipment	9
4.2.4 RSW Setup and Validation	11
4.2.5 RSW Weld Certification	13
4.2.6 RSW Launch	15
4.2.7 RSW Design Tables and Drawings	16
4.2.8 Weld Timer Maintenance Schedules	20
4.2.9 Support and Services	21
4.3 Procedure	22
4.4 Process control	22
4.5 Inspection requirements	22
4.6 Re-process	22
5 SPECIAL REQUIREMENTS	23
6 QUALITY Concerns Regarding Expulsion Reduction	24
7 TEST EQUIPMENT	24
8 SAFETY PRECAUTIONS	24
9 APPROVED SOURCE LIST	24

1 GENERAL

The intent of this document is to provide joining direction for the launch of the program.

1.1 Purpose

This specification is to supplement the various welding design and process standards already in place. It specifies the actual procedures, deliverables, and certifications to ensure a smooth product launch in Body-In-White facilities.

1.2 Coverage of this Standard (Applicability) and Limitations on Usage)

While FCA uses many different welding processes, this standard will only address Resistance Welding

2 REFERENCES

Table 1 - References

Document Number	Shield/Design ator (if applicable)	Document Title	downloadable for suppliers from beStandard
CEP.00023/02		DESIGN – METALS JOINING – SPOT WELDING	Y
PS.50004/01	S	WELDING – RESISTANCE SPOT	Y
PS-A0360		ALUMINUM RESISTANCE SPOT WELDING SPECIFICATION	Y
CS-A0210		RESISTANCE WELD GUN CONSTRUCTION SPECIFICATION	Y
CS-A0211		WELD CAP DRESSER WITH INTEGRATED CHIP COLLECTION SPECIFICATION	Y
NFPA 70 [Note4]		NATIONAL FIRE PROTECTION ASSOCIATION	N
PS-11168		BODY AND EXTERIOR LAUNCH CERTIFICATION PROCESS	Y
PS-A0212		FUSION WELD LAUNCH PROCEDURES	Y
CS-A0208		INVERTER STYLE RESISTANCE WELDING CONTROLLER	Y
ME 200-22		LINE INTEGRATOR REQUIREMENTS	N
AWS f2.3-01 [Note2]		SPECIFICATION FOR USE AND PERFORMANCE OF TRANSPARENT WELDING CURTAINS	N
MQGR-30 [Note3]		WELDING/STRUCTURAL ADHESIVE OPERATIONS – MFG. & ASSEMBLY PLANTS	N
MS Excel Document [Note1]		EXAMPLE OF PROJECT WELD MATRIX.XLS	N
MS Excel Document [Note1]		EXAMPLE OF EXPULSION REDUCTION TRACKING SHEET.XLS	N
MS PowerPoint [Note1]		WELD EXPULSION REDUCTION PLAN OEM.PPT	N
MS Excel Document [Note1]		EXAMPLE OF WELD VERIFICATION TRACKING MATRIX.XLS	N
MS Excel Document [Note1]		EXAMPLE OF WELD DESTRUCT TRACKING MATRIX.XLS	N
MS Excel Document [Note1]		RESISTANCE SPOT WELD VERIFICATION BUY OFF.XLS	N

[Note 1] Document is available through FCA ME Welding CoC group or the specifying authority

[Note 2] Document can be found at <http://pubs.aws.org/>

[Note 3] Document can be found at <http://sctcda06.wk.dcx.com/852574CD005638A8.nsf/> A valid FCA user ID is required to access MQGR-30.

[Note 4] Document can be found at <http://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards?mode=code&code=70>

3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS

- OEM – Original Equipment Manufacturer
- RSW – Resistance Spot Welding
- GMT – Governing Metal Thickness
- KVA – Kilo Volt-Amperes
- MFDC – Mid-Frequency Direct Current
- CoC – Center of Competency
- SCR – Silicon Controlled Rectifier
- FCA ME – Fiat Chrysler Automobiles Manufacturing Engineering
- WCM – World Class Manufacturing
- PPSL - Program Preferred Source List
- HMI – Human Machine Interface
- AHSS – Advanced High Strength Steel

4 PROCESS AND PROCESS CONTROL REQUIREMENTS

4.1 Condition of the part prior to processing

Not Applicable.

4.2 Resistance Spot Welding

4.2.1 RSW Design

The following shall apply to resistance spot welding.

Metal Thickness

Determination of governing metal thickness (GMT) (Refer to **Table 2 Process and Design References**):

- For 2t stack-ups, the governing metal thickness (GMT) is the thinnest sheet. If the two sheets are the same thickness then that thickness is the GMT
- For 3t stack-ups, there are two governing metal thicknesses GMT1 and GMT2, one for each faying surface. GMT1 is the thinnest of sheets 1 and 2. GMT2 is the thinnest of sheets 2 and 3. From the recommended welding set-up tables, compare the settings for both GMT's and use the worst case to determine the appropriate welding parameters and adaptor size. If thinnest sheet is located in the middle, the thinnest outside sheet shall be used for GMT on the schedule selection charts

Laminated Steel

When welding laminated steel consider the sum of the two steel sheets comprising the laminated steel as a single solid sheet. Laminated steel may require different weld set-ups than shown in Table 4 (Design References in **Section 0** of this standard).

Weld Schedule Development

Welding time, current and force are to be implemented from Table 4 Design References found in **Section 0** in this standard. **Direction to use cycles or milliseconds shall be given by FCA ME Joining CoC.**

Weld Cycle Time

All welds shall be processed for 1 second cycle time (60 cycles at 60 Hz). This cycle time is measured from the start of "Squeeze" until the end of "Hold". The 60 cycles of weld time does not include robot moves, gun open/close, etc. This weld cycle time shall also be utilized in transformer sizing calculations. All weld times identified in this specification are shown in cycles. These times may be required to be converted to milliseconds, depending on the assembly plant.

Weld Force and Adaptor Sizing

The system integrator shall be responsible for sizing all weld guns utilizing the information supplied in Table 4 Design References found in Section 4.2.7 of this specification. Weld adaptor sizing shall be 3/4" weld caps for GMT's > 1.50 mm, 5/8" caps for GMT's ≤ 1.50 mm. For aluminum welding, see **PS-A0360 Aluminum Resistance Spot Welding Specification** for additional information.

Laminated Leaf Shunts

Leaf shunts shall be sized correctly for thermal loading and located at the optimum position such that no evidence of physical damage shall occur when the weld gun closes and current passes through (Further requirements can be found in **CS-A0210 Resistance Weld Gun Construction Specification**).

Transformer Sizing

Size of the transformer shall be calculated by the system integrator utilizing the worst case stack-up on the application. Reference **CS-A0210** for additional details and information regarding transformer sizing.

Transformer Grounding

All non-current carrying metallic components of the DC Power supply shall be bonded as required by Article 250 of **NFPA 70** national electric code.

Transformer Cooling

See **CS-A0210 Resistance Weld Gun Construction** Specification for transformer cooling requirements.

Weld Gun Design

All weld guns shall be designed and/or selected in accordance with **CS-A0210 Resistance Weld Gun Construction** Specification.

The system integrator shall design and integrate all weld guns in accordance with the robot manufacturing requirements. Mounting hardware shall be designed and provided by the system integrator.

MFDC Servo Trans-guns

Where applicable, the provided RSW systems shall use MFDC servo trans-guns.

Weld Gun Selection

The system integrator shall be responsible for selecting a suitable weld gun from the FCA library prior to tooling design. The weld gun design shall be confirmed during the tooling design process. Any process that may propose a new gun design shall require a signed deviation from the FCA Weld Gun Group and ME Joining CoC.

Weld Gun Requirement

All weld guns shall be able to fit into a standard tip dresser.

Multiple Weld Guns

Where dual weld guns are utilized, separate transformers and weld controls shall be used for each gun. The OEM shall be permitted to use a cascade firing a single weld control with a multiple SCR. A written deviation shall be provided to and approved by FCA ME Joining CoC.

Hard Tool Guns

Where hard tool weld guns are utilized, individual transformers shall be used for each weld gun and controlled individually. The OEM shall be permitted to use a cascade firing a single weld control with a multiple SCR. A written deviation shall be provided to and approved by the FCA ME Joining CoC.

Weld Controller

Weld controllers shall be sized for the application and have the ability to perform a minimum of 10 million welds. All weld controllers shall be supplied by FCA unless otherwise specified.

Weld output cables

Weld output cables shall be sized to the maximum output of the inverter.

Tip Dressers

The system integrator shall design, purchase, supply and integrate new tip dressers, cutters and control equipment for all new RSW weld systems (robotic, pedestal, and hard-tool if applicable). Tip dress shall exhibit a method of detecting motor stalling. All tip dressers shall be able to achieve +/- 50 lbs. from nominal tip dress force without stalling.

All tip dressers shall be driven by electric servo motors; if another method is preferred it shall require a written buyoff from FCA ME Process Engineer and Joining CoC. Kit Jobs should utilize TEFC electric motors if the existing equipment's motor type is electric TEFC to ensure uniformity throughout the production line. These shall comply with the current manufacturer and be listed on current PPSL.

If the use of pneumatic tip dressing motors are deemed to be required, a multi-flute cutter shall be implemented (Min of 3 blade cutter); It must be noted that the use of these pneumatic motors will require FCA ME approval before use. All non-servo tip dress motors shall be controlled by the PLC and conform to **CS-A0211 Weld Cap Dresser with Integrated Chip Collection Specification**.

Weld guns are not permitted to share a tip dress unit.

Tip Dress Process

The system integrator shall incorporate the following procedure to the tip dress process:

- Water shall be present before the caps enter the tip dresser. This shall be monitored to ensure that the adaptor does not get damaged.
After dress is completed a burn-in schedule (schedule 255) shall be fired.
- Schedule shall be 18 weld cycles, 10.0kA current and 480lbs of weld force.
- All associated cycle time as a result of the tip dress process is required to be accounted for within the robots overall cycle time.

Tip Dress Stands and Mounts

Tip dress stands and end effector mounted units shall be reviewed and approved in accordance with the dresser equipment manufacturer's requirements. All end effector mounted servo tip dressers shall be insulated from the material handler. Tip dress stands and mounts are to be welded back to a common and stable point.

Tip Dress equipment shall be mounted for easy access for maintenance and inspection. Installation location shall allow line of sight to the uppermost cutter face and maintenance operations to be performed from the floor. The equipment shall be mounted vertically when installed below 32 inches. The equipment may be mounted horizontally or vertically when installed above 32 inches and below 48 inches. The equipment shall be mounted vertically when installed above 48 inches. Any design or installation outside of these parameters must be approved by the FCA ME Joining CoC. If mounted horizontally over a height of 48 inches a step or platform shall be provided to allow line of sight access to the uppermost cutter face.



Figure 1 - 'Tip Dresser Mounting Configuration and Direction

Pedestal Weld Gun Tip Dress

All pedestal mounted weld guns shall utilize dump mounted tip dressers. If areas concerning floor space require a special process it shall be approved by FCA ME Process Engineer and Joining CoC. The tip dress dump shall be capable of being completely stabilized within 1 second of the dump extended.

Weld Caps

Weld Caps shall be FCA Standard weld caps and sourced from the program preferred supplier with the following characteristics:

- Material shall be Class 1 zirconium
- 5/8" & 3/4" ball nose with a 4.8 mm face
- Specialty weld cap applications shall be approved by the FCA ME Joining CoC
- See **PS-A0360 Aluminum Resistance Spot Welding Specification** for weld cap types when welding aluminum.

Chip Collection

The system integrator shall provide chip collection units for all tip dressers with vacuum function to collect copper shavings from the tip dressers. This collection system shall be in accordance with FCA WCM standards. An OEM solution of failure detection for the vacuum system shall be incorporated in the tip dresser functionality.

Tip Dress and Cap Change Request

An HMI feature shall be included to manually send each individual robot, all robots by station, and by auto zone to tip dress, maintenance position and cap change when necessary.

Weld Cap Trays

See **CS-A0210 Resistance Weld Gun Construction** Specification for weld cap tray requirements.

Automated Cap Changers

When half-length weld caps are required, or it is determined by the Joining CoC that a weld gun will require more frequent cap changes than a full shift of production, the system integrator shall provide an automated weld cap changer. See **CS-A0210 Resistance Weld Gun Construction** Specification for additional requirements.

4.2.2 RSW Documentation

The following section shall apply to resistance spot welding documentation.

Weld Spot Matrix

The system integrator shall assign one person full time from the beginning of tool design until V1 launch of the vehicle associated with this program. This person shall be responsible for downloading all weld files, inputting required fields and verifying accuracy of information. All changes in documentation shall be highlighted and summarized after each updated release. The spot matrix creation begins at the Engineering Step 3 release. Time frame from this release to a usable weld spot matrix shall be two weeks. Each subsequent milestone release shall require an updated weld spot matrix within one week of release date (File name: Example of Project Weld Matrix.xls).

Create and Maintain Weld Expulsion Tracking Matrix

The system integrator shall rate, track and update all robots for expulsion reduction (File names: Example of Expulsion Reduction Tracking Sheet.xls, Weld Expulsion Reduction Plan OEM.ppt). Weld Expulsion Tracking Sheet shall be updated and returned to Lead FCA ME Joining CoC Engineer weekly.

Weld Gun Verification Sheets

The system integrator is to update through weld gun verification process a weld verification tracking matrix (file name: Example of Weld Verification Tracking Matrix.xls).

Weld Certification Documentation

The system integrator is to maintain weld certification documentation through V1 (file name: Example of Weld Destruct Tracking Matrix.xls).

Transformer Identification

See **CS-A0210 Resistance Weld Gun Construction** Specification for transformer identification requirements.

Part Number Placard

The system integrator shall provide an 11 x 17 laminated placard at each station containing a graphic with part numbers for use by plant trades in the replacement and maintenance of weld guns at each welding station.

Weld Gun Motor Labeling

Oil resistant labels shall be mounted on each servomotor. Reference **Section 5.37** of **CS-A0210 Resistance Weld Gun Construction** Specification.

Weld Gun Maintenance

The weld gun supplier or equipment integrator shall provide a maintenance manual that will provide enough detail to enable plant maintenance personnel to repair service and order replacement parts via the FCA NPM code system.

4.2.3 RSW Build & Equipment

The following section shall apply to resistance spot welding and equipment.

Component Tryout

The system integrator is responsible to tryout all applicable components, prior to shipping to the FCA plant.

Approved Welding Equipment

All welding equipment components and devices provided by the suppliers shall be on the approved PPSL for the program.

Weld Schedule Tryout

The system integrator is to fire every weld schedule programmed in the robot to ensure no parameter has been entered incorrectly. All schedules shall be validated by a calibrated current meter.

Weld Schedule Download

The system integrator is responsible for inputting all schedule and force callouts into the robot programs. Weld schedules and weld forces are to be setup up for one schedule per weld spot and shall be sequential with the robot sequence. The system integrator is also responsible to enter in all weld schedules into the appropriate weld controllers. The IP address for each weld control also is required to be input with initial weld schedule download (See **Table 10 Maintenance Weld Schedules**).

Weld Controller Self-diagnostic

All welding controllers shall have self-diagnostic functions for all possible failure modes. The controllers shall be able to communicate the fault messages to the robot teach pendants, HMI screens and the program weld network (where applicable).

Teach Pendant Control

All programming and diagnostics are to be through the robot teaching pendant on robot controlled weld controllers. This is not applicable to the hard tool weld controllers.

Weld Power Supply Location

The default location of the welding power supply (Resistance, fusion and projection welding) shall be on the top of the robot controller. If this location needs to be different, FCA ME Process Engineering shall review and approve as required.

Weld Controller Software Update

The system integrator is responsible to update all welding controller software for the duration of the program.

Weld Software Approval

All welding related software updates or upgrades shall be approved and coordinated with FCA ME Process Engineering and Joining CoC prior to implementation.

Primary Quick Disconnect

The system integrator shall supply a primary quick disconnect at the weld transformer for all new servo weld guns. Weld guns >165 KVA shall require hard wiring the primary.

Dry Cycle Weld Schedule Setup

Dry Cycle mode shall utilize production weld schedules to simulate real production cycle times. A Weld ON and Weld OFF selection shall be set remotely from the HMI to disable the weld mode in order to prevent weld equipment damage during machine debug and verification activities. At no time shall the Weld Mode become disabled in Automatic Mode with Dry Cycle disabled. The Dry Cycle Weld OFF selection shall only become enabled when the Dry Cycle Mode is preselected.

Weld Cap Change

The system integrator shall install new weld caps prior to programming the weld gun to metal, prior to running weld certification parts, and after dry cycles are run.

Tip Dress Setup

The following parameters shall be used as a baseline tip dress routine for steel applications only:

- Tip dress frequency shall not exceed 250 welds with a 2 amp/weld boost for each weld
- Tip dress force shall be 300lbs unless the cap manufacturer specifies otherwise
- Tip dress time shall be set to 0.75 seconds (45 cycles) or until the cap face is back to original face, size and cap profile
- Max number of tip dresses shall be set to 40 on standard caps and 20 on any short cap applications; this number can be adjusted when production begins to optimize the cap life per robot.
- The system integrator is responsible to ensure weld cap life expectancy to be equal to or greater than 1 shift of production (8 hours or 10 hours depending on the assembly plant).

Cap location

The spot welds placed on the part shall be programmed without current shunts and be programmed with the stationary cap taught touching and perpendicular to the work surface.

Edge Welds

Edge welds are not permitted. Welds shall be centered across the width of any narrow weld flanges. Outside diameter of weld cap shall be clear of any upstanding flange by a minimum of 2 mm.

Dress Routing of Cables and Hoses

The system integrator shall route all welding cables, air hoses, and water hoses to eliminate any rubbing, pinching, or snagging.

Cap Seating

Following a cap change, a service robot program shall be run to close/open the gun three (3) times to properly seat the caps. Weld guns with 5/8" weld caps shall be seated with 950 lbs of force and 3/4" weld caps shall be seated with 1,200 lbs of force, unless otherwise specified by the Joining CoC. Weld caps shall be seated before enabling water flow.

Secondary Feedback and Adaptive Welding Set-up

All welding systems shall have secondary feedback and adaptive welding capabilities.

System integrator is responsible for the cabling from the weld controller to the adaptors of the secondary feedback wires. This is the purchase of the weld control supplier's preferred cables, plus the actual installation of the cables. The system integrator is also responsible for ensuring the master offset has been completed and documented on the weld timer sheet.

Steppers shall be set-up the same as constant current welding.

4.2.4 RSW Setup and Validation

The following section shall apply to resistance spot welding setup and validation.

Weld Certification

New weld caps shall be used for weld certifications.

Weld Verification Requirement

The welding verification shall be conducted expulsion-free. Prior to the certification, steps 1 thru 10 below in welding expulsion reduction shall be completed. They are:

1. Tip-to-tip alignment
2. Gun-to-metal alignment
3. Tip offset programming (if applicable)
4. Good part fit-up, gun force verified
5. Sealant/adhesive contamination free
6. Tip dressing verified
7. Weld path touched up
8. Verify the part metal stack-up
9. Verify the weld gun squeeze time
10. Verify weld data

For explanation on this activity see "Example of Weld Expulsion Reduction Plan OEM.ppt".

Weld Current Calibration

The system integrator is responsible for ensuring that the weld current is within 300 amps of desired programmed weld currents. The system integrator shall validate with the following process:

- Biddle meter all connections in the secondary of the weld gun
- Program a default weld schedule (Schedule 251):
 - 10 cycles squeeze
 - 10 cycles weld time
 - 25% amps (% I mode)
 - 5 cycles hold
 - 480 lbs. of force
 - Weld schedule to be fired tip to tip with calibrated current meter Integrator to verify and document current on timer sheet
- No loose connections (no sparking present at any connection)
- Fire all weld schedules used in the process to confirm information has been entered correctly (+/- 300 amps of programmed weld current)
- Current meter matches weld timer current reading (proper turns ratio & timer calibration)
- Meter/weld control to verify no half cycling
- Verify normal gun operation (close tip to tip, weld force, no shunting in the gun, etc.)
- Record nominal C-Factor to be used as a baseline

Weld Schedule Adjustments

The system integrator shall be responsible for all discrepant weld conditions reported from all weld certification processes. Weld schedule updates shall be reported to the FCA ME Joining CoC and updated on the program Weld Spot Matrix.

C-factor Setup

The system integrator shall input C-factors for the burn-in weld schedule (Schedule 255) before any level of product is run through the system. The limits shall be set to -20% (low) / +20% (high) from nominal reading and shall be programmed as "Faults". These limits shall be set utilizing the automated tip dressing process to dress the weld caps, execute five manual welds and record the measured C-Factor for each weld. Average the five C-Factor measurements collected and set the 20% window around that average C-Factor.

Actual welding schedules shall be programmed as "Alerts". Joining CoC will advise on the methodology of setting C-Factor limits for welding schedules. Nominal values shall be established from V1 part runs.

Current Limit Setup

The system integrator shall input Current Limits in the global setup for each weld controller, there shall be no Current Limit windows inserted into individual weld schedules. The global setup Current Limits shall be set to -20% (low) / +20% (high) and shall be programmed as "Faults".

Robots shall be setup to allow a maximum of one (1) auto retry weld to reduce nuisance line stoppages.

Weld Gun Force

The system integrator shall measure gun forces at all required welding forces to validate the process intent which includes all weld forces, tip dress force and max gun force. All tested forces are to be in accordance with **CS-A0210 Resistance Weld Gun Construction** Specification requirements. If a force falls outside of the range, it is the responsibility of the system integrator to re-calibrate the weld gun.

Validation of Weld Gun Cooling

When a weld gun is deemed ready to weld, the system integrator shall record ambient temperature of weld adaptors with a calibrated thermal device. The integrator is to fire 5 welds within a 30 second time frame and reshoot the temperature. Temperature shall return within 10 degrees of recorded ambient temperature within 10 seconds of the last weld fired. The gun manufacturer shall be responsible for any discrepant cooling conditions. If the weld gun does not demonstrate the above mentioned cooling, water flow design requirements shall be in accordance with **CS-A0210 Resistance Weld Gun Construction** Specification. Each path shall be verified with an in-line flow meter with all circuits open

Water Tube Verification

The system integrator shall verify that all water tubes in accordance with **CS-A0210 Resistance Weld Gun Construction** Specification. The weld gun manufacturer shall be responsible for any water tubes that fall outside the requirements.

Tip Dress Verification

The system integrator shall verify that weld caps are normal to the tip dress cutter blade and color weld cap face black to ensure that the tip dress parameters return face to nominal (4.8mm) and no color is remaining.

Tip Alignment Verification

All weld guns are to be closed and tip to tip normality is to be verified with an alignment tool. Any alignment issues are the responsibility of the gun manufacturer to repair. A report depicting any off normal situations shall be reviewed with FCA ME Process Engineering and signed off prior to implementation.

4.2.5 RSW Weld Certification

The following shall apply to resistance spot welding certification.

Weld Certification Cost

The system integrator shall be required to perform and provide cost for the following actions based on the weld quantities listed below:

- All weld destruct methods and requirements are specified in the **PS.50004/01** specification.
- Every weld on the spot matrix for assigned sub shall be identified and recorded.
- System integrator is to provide FCA ME Joining CoC a completed weld destruct matrix for approval and weld certification.
- System integrator shall be responsible for all shipping arrangements and costs to off-site teardown (if applicable)
 - Up to 200 welds – documentation will be provided to FCA within 24 hours.
 - 201 to 499 welds – documentation will be provided to FCA within 36 hours.
 - 500 to 1000 welds – documentation will be provided to FCA within 48 hours.
 - 1001 to 2000 welds – documentation will be provided to FCA within 72 hours.
 - 2001 to 4000 welds – documentation will be provided to FCA within 96 hours.

Build Phase Teardown Frequency

The system integrator shall be required to follow all these actions for the following build phase and amount of teardowns required at each phase:

- TTO Build – 3 weld teardowns
- VPA Build – 1 weld teardown
- VPB Build – 1 weld teardown
- VPC Build – 1 weld teardown
- PS Build – 1 weld teardown
- Post V1 launch - 1 weld teardown reaching 98% weld effectiveness
- Any additional teardowns due to unacceptable welding that exceeds the predetermined amounts set above shall be the system integrator's responsibility

Assumptions

Assumed amount of welding for purposes of this quotation:

- Rear Floor - 200 welds
- Rear Rails RH and LH - 300 welds
- Engine Box - 275 welds
- Front Floor - 250 welds
- Underbody Framer - 300 welds
- Underbody Respot - 650 welds
- Quarter Inner RH and LH - 200 welds
- Aperture Outer RH and LH - 600 welds
- Framer 1 - 100 welds
- Framer 2 - 250 welds
- ARS - 375 welds
- Roof Line - 50 welds
- Front Doors RH and LH - 150 welds
- Rear Doors RH and LH - 125 welds
- Fender - 50 welds
- Hood - 25 welds

Multiple Process Applications

The system integrator shall be responsible for identifying which lane a part is processed through by a unique weld spot identifier. The identifier needs to be easily noticeable when presented in the pull off station. See **Figure 2 - Multiple Line Identifiers Example** for reference.

- A unique identifier is required every time a part is (re)introduced into lane "A". For example, every robot in lane "A" shall require a unique identifier if the ability to change lanes is present directly downstream of the robot.
- Identifier shall be a Manufacturing weld only. System integrator shall provide FCA ME Process Engineer with location so a spot number can be released for all documentation updates.
- Pull off stations need to be present for all multiple processes in a way that all welds can be inspected before being closed out.
- Pull off stations shall be designed to individually request an assembly from any lane.
- When a pull off request is active, the assembly cannot cross over to the opposite lane at any point during the process of the predetermined line.

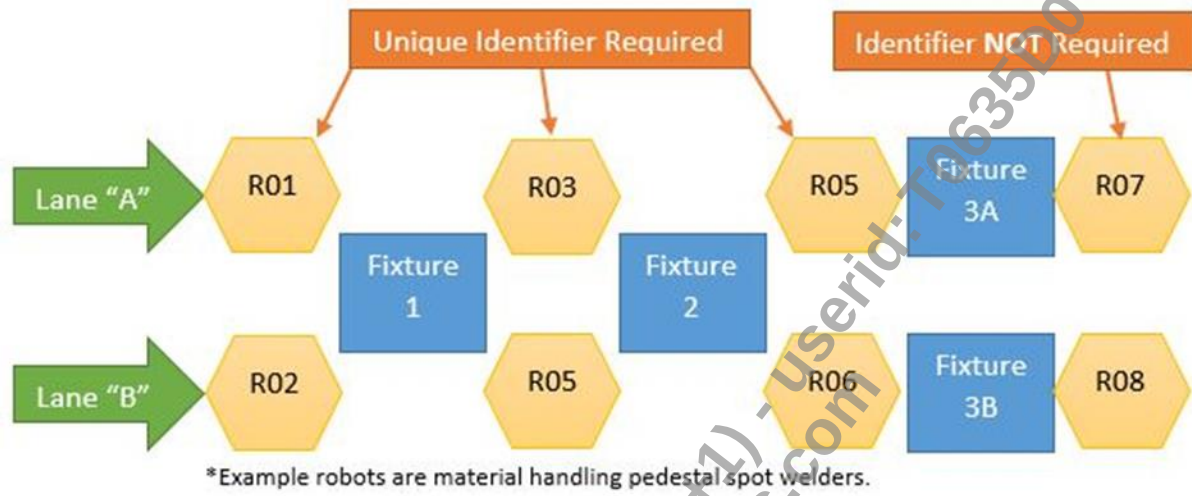


Figure 2 - Multiple Line Identifiers Example

4.2.6 RSW Launch

The following section shall apply to resistance spot welding launch.

Launch Weld Support

The system integrator shall have a minimum of one qualified weld engineer for all part builds assigned to each production plant. Eight hours minimum of welding support for all builds plus 24 hour availability are required when needed. Support shall continue for three months post V1 launch for expulsion reduction activities and cap life optimization. If expulsion reduction and cap life optimization activities are not completed following three months post V1, the system integrator shall be required to continue to provide support until these activities are completed.

Post V1 Weld Teardown

The system integrator shall achieve 98% weld conformance on a Post V1 weld teardown. If additional units are required to achieve 98% conformance past the one required corporate teardown, the system integrator shall be responsible for providing any additional costs for teardown manpower.

Expulsion Reduction Activities

The system integrator is to implement the FCA Weld Expulsion Reduction Plan OEM.ppt. No expulsion shall be allowed to leave any welding cell and levels of overall expulsion throughout each system shall be less than 15% weld expulsion to deem task complete. Expulsion reduction does not allow for any exceptions for cycle time or nugget requirements. Recording the adjustments made for eliminating weld expulsion and updating the program Weld Spot Matrix shall be the responsibility of the system integrator.

Parts Build Consumables

The system integrator shall provide and maintain parts build consumables. Reference ME BIW 200-xx Line Integrator Requirements Specification, Section 11 for required quantities of parts build consumables.

Weld Verification Buy Off

The system integrator shall be responsible for all resistance spot welding equipment until the tool has officially been bought off by FCA ME Joining CoC.

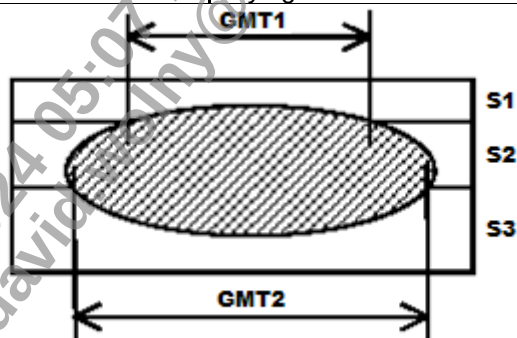
- Use the "Resistance Spot Weld Verification Buy Off.xls" checklist for additional information
- Every item on this checklist shall be completed by the System integrator and bought off by FCA ME Joining CoC for the tool to be considered completed and in accordance with this specification.

4.2.7 RSW Design Tables and Drawings

The following tables and drawings shall be used in association with the RSW Design section.

Table 3 - Three thickness minimum weld size requirements – Spot welding coated and uncoated (bare) low carbon (mild) steels, High strength steels, or advanced high strength steels
(1)

Total Metal Thickness, mm	Minimum Required Weld Size mm (2) (3)
2.67 - 8.0	There are two minimum required weld sizes for three thickness welds, GMT1 and GMT2 as shown. The two minimum required weld sizes shall be evaluated separately based on the GMT in Table 1 for each faying surface.



NOTE 1: Applicable to all classes of welds.

NOTE 2: An etched cross section of the weld should show 20% to 90% penetration.

NOTE 3: See Section 2.6.1 of **PS.50004/01** for instruction on measurement / evaluation of weld size.

NOTES FOR TABLE 4

- (1) The weld set-ups indicate weld time, pulsation, weld current force and tip size. The weld set-ups assume identical tips on any given weld gun. All weld tips are assumed 4.8 mm face class 1 chrome zirconium. Adjustment of any portion of the set-up is permissible for differing thick-thin combinations, differing fit-up conditions, and other situations that affect the specific welder being adjusted.
- (2) Laminated Steels may require different weld set-ups than shown in this table.
- (3) When sheet thickness ratio exceeds the limitations in Tables 12 and 13, contact FCA ME Joining CoC for recommended weld set-ups.
- (4) Total weld time includes squeeze time, pulse time, cool time between pulses and hold time.
- (5) Adaptor size will change from 5/8" to 3/4" at the governing thickness of 1.50 mm.
- (6) Weld forces are considered acceptable one force range higher or lower than recommended force when approved by the Joining CoC.
- (7) Weld set-ups for coated materials are based on 55g coating weight. Heavier coating weights may require increased amps.
- (8) These weld tables should not be used as the only means of sizing necessary weld equipment

**Table 4 - Recommended Welding Set-Up:
Mild Steel <50 KSI (350 MPa)-
Two Or Three Thickness - (1) (2) (3)**

GMT (3) mm	TOTAL TIME (4) (60 Hz cycles)	Pulse Time (60 Hz cycles)	Cool Time (60 Hz cycles)	Number of Pulses	Kilo Amps (8)	Hold Time (60 Hz cycles)	Weld Force Mild Steel in Kg (lbs) (4)(5)
.60 - 1.02	26	18	0	1	8.5	2	218 (480)
1.03 - 1.23	34	7	1	3	8.5	5	304 (670)
1.24 - 1.49	34	7	1	3	9.0	5	304 (670)
1.50 - 1.64	42	7	1	4	9.0	5	431 (950)
1.65 - 1.79	50	7	1	5	9.5	5	431 (950)
1.80 - 2.09	50	7	1	5	10.0	5	544 (1200)
2.10 - 2.45	50	7	1	5	10.0	5	544 (1200)
2.46 - 2.69	58	7	1	6	10.5	5	544 (1200)
2.70 - 3.07	58	7	1	6	10.5	5	753 (1660)
3.08 - 3.20	58	7	1	6	11.0	5	753 (1660)

**HSS 50 KSI (350 MPa) ≤ TS ≤ 73 KSI (500MPa)-
TWO OR THREE THICKNESS - (1) (2) (3)**

GMT (3) mm	TOTAL TIME (4) (60 Hz cycles)	Pulse Time (60 Hz cycles)	Cool Time (60 Hz cycles)	Number of Pulses	Kilo Amps (8)	Hold Time (60 Hz cycles)	Weld Force HSS in Kg (lbs) (4)(5)
.60 - .89	26	18	0	1	8.5	2	218 (480)
.90 - 1.23	34	7	1	3	9.0	5	304 (670)
1.24 - 1.49	34	7	1	3	9.0	5	431 (950)
1.50 - 1.64	50	7	1	5	9.5	5	431 (950)
1.65 - 1.79	50	7	1	5	9.5	5	544 (1200)
1.80 - 2.09	58	7	1	6	10.0	5	544 (1200)
2.10 - 2.45	58	7	1	6	10.5	5	544 (1200)
2.46 - 2.69	58	7	1	6	10.5	5	753 (1660)
2.70 - 3.07	58	7	1	6	11.0	5	753 (1660)
3.08 - 3.20	60	9	1	5	11.0	5	753 (1660)

AHSS >73 KSI (500 MPa)-

Table 4 - Recommended Welding Set-Up:

Mild Steel <50 KSI (350 MPa)-
Two Or Three Thickness - (1) (2) (3)
TWO OR THREE THICKNESS - (1) (2) (3)

GMT (3) mm	TOTAL TIME (4) (60 Hz cycles)	Pulse Time (60 Hz cycles)	Cool Time (60 Hz cycles)	Number of Pulses	Kilo Amps (8)	Hold Time (60 Hz cycles)	Weld Force AHSS in Kg (lbs) (4)(5)
.60 - .89	28	8	1	2	8.0	5	218 (480)
.90 - 1.02	34	7	1	3	8.0	5	304 (670)
1.03 - 1.25	42	7	1	4	8.5	5	431 (950)
1.26 - 1.49	50	4	1	8	9.0	5	431 (950)
1.50 - 1.84	55	4	1	9	9.0	5	544 (1200)
1.85 - 2.09	55	4	1	9	9.5	5	544 (1200)
2.10 - 2.45	60	4	1	10	9.5	5	753 (1660)
2.46 - 2.69	60	4	1	10	10.0	5	753 (1660)
2.70 - 3.07	60	4	1	10	10.5	5	753 (1660)
3.08 - 3.20	60	4	1	10	11.0	5	753 (1660)

Table 5 - Minimum requirements for equipment set-up verification and weld certification

Stage	Weld Size (1) (2)	Weld Gun Verification Procedure	Weld Destruct Assembly	Non-Destructive Verification (refer to Table 12)	Visual Check of Weld Equipment
At initial machine set-up (Prototype, Pilot and Production)	Weld Size ≥ Min. PS.50004/1	Required	Required	N/A	N/A
After major maintenance (3), set-up changes, or changes to steel supplier or grade (Prototype, Pilot and Production)	Weld Size ≥ Min. PS.50004/1	N/A	Required (Select 1 of 2 options)		N/A
During prototype & pilot build	Weld Size ≥ Min. PS.50004/1	N/A	N/A	· All Safety welds on all scheduled impact and durability vehicles. · All welds on every 25th assembly (4)	N/A

Table 5 - Minimum requirements for equipment set-up verification and weld certification

Stage	Weld Size	Weld Gun Verification Procedure	Weld Destruct Assembly	Non-Destructive Verification (refer to Table 12)	Visual Check of Weld Equipment
	(1) (2)			PS.11168	
During production	Weld Size $\geq$ Min. PS.50004/1	N/A	- BIW assy: Required at V1 launch and mid model year. - Stamping, Power Train, Component, Satellite Plants and all suppliers: Required every 2 weeks (5). MQGR-30	Plant or supplier to define requirements in control plan.	Required two times per operating shift (safety welds only) – EM-10001 <S>, EM-10002 <S>, EM-10003 <S>.

General A sampling plan is required to contain all discrepant locally applied welds (F).

NOTE 1: Refer to Section 4 in **PS.50004/01** for complete Weld and Inspection Requirements.

NOTE 2: Ref. S.11168 for certification requirements for launch build events.

NOTE 3: Details of what is considered major maintenance are outlined in MQGR-30 Table 4

NOTE 4: Non-destructive ultrasonic inspection of the assembly may be replaced with weld destruct inspection.

NOTE 5: Destructive testing can be reduced to once per month if ultrasonic testing is used.

NOTE 6: FCA Group North American Assembly, Stamping and Component Plants should refer to **MQGR-30**.

4.2.8 Weld Timer Maintenance Schedules

Table **Error! Reference source not found.** is a list of the dedicated weld schedules to be allocated for maintenance purposes. These shall be preloaded as defaults by the weld controller manufacturer.

Table 6 - Maintenance Weld Schedules

Schedule #255 – Weld tip burn-in/current calibration/tip resistance check. This schedule shall be set up with 5 cycles of squeeze time, 20 cycles of weld time with 10.0kA weld current in constant current mode and 5 cycles of hold time. Stepper function shall be required to be on.

Schedule #254 – Tip dress function/stepper reset. This schedule shall simply reset the stepper to step 2 and shall contain **NO** heat commands. The schedule shall be set up with 75 cycles of squeeze time and 5 cycles of hold time as a default.

Schedule #253 – Weld gun force check/tip seating. This schedule shall contain **NO** heat commands. The schedule shall be set up with 30 cycles of squeeze time and 30 cycles of hold time as a default.

Schedule #252 – Dry cycle/initial gun test. This schedule shall be programmed for 30 cycles of squeeze time, 15% heat at 10 cycles and 5 cycles of hold time in AVC mode and no stepper.

Schedule #251 – Weld current calibration. 10 cycles of squeeze time, 10 cycles of weld time at 25% secondary, 5 cycles of hold time under 480 lbs of force.

Schedule #250 – Reserved for future use

Table 7 - Most Commonly Used Steel Types

Mild Steel	Tensile strength < 350 Mpa (50 Ksi)
	MS.50002 CRxx 60/60
High Strength Steel (HSS)	350 Mpa (50 Ksi) ≤ Tensile strength ≤ 500 Mpa (73 Ksi)
	MS.50002 LAC380Y450T
Advanced High Strength Steel (AHSS)	Tensile strength > 500 Mpa (73 Ksi)
	MS.50002 DPC700Y980T

4.2.9 Support and Services

Breakdown Plan

The system integrator shall provide a breakdown plan on the resources, such as work force, for weld process development, weld certification, and on-site support during different launch phases.

Machinery–FMEA's

Welding equipment manufacturers shall provide Machinery–FMEA's for all major welding equipment. The M-FMEA shall be reviewed by FCA ME Process Engineering and Joining CoC. The revision of the existing M-FMEA may be required.

Process-FMEA Meeting

The system integrator and welding equipment manufacturers participate in a minimum of three (3) meetings for Process–FMEA, if it is required.

Operation Instructions and Maintenance Documentation

Welding equipment manufacturer shall provide the operation instructions and maintenance documentation, including machine ledgers. One hard copy and one electric copy (flashdrive) are required per cell.

Documentation

The system integrator shall provide an electronic copy of following documentation:

- Bill of materials (BOM)
- Installation drawings
- Mechanical drawings
- Wiring diagrams (WDs)
- Communication (Devicenet and Ethernet) files

Maintenance Tools

Welding equipment manufacturer shall provide the mechanical and electrical maintenance tools if they are special.

Training Requirements and Plan

Welding equipment manufacturer shall provide the training requirements and plan for plant maintenance personnel.

Spare Parts List Format

All tier-2 system integrators shall provide recommended spare part lists using FCA defined template format.

Try-out Support

During system and equipment tryouts all Tier-2 system integrators shall provide on-site support.

Buy-off Support

After system buyoff, the system integrator and all Tier-2 system integrators shall provide 24-hour, including weekends and holidays, on-call services. The "hot line" phone numbers and contact persons shall be posted on site. The required response time is two hours or less. The service fees shall be based on the mutual agreements between the System integrator and FCA.

4.3 Procedure

Not Applicable.

4.4 Process control

Not Applicable.

4.5 Inspection requirements

Not Applicable.

4.6 Re-process

Each plant is required to have a designated repair area or areas for units with discrepant resistance welds. The plant will also be responsible for providing the weld reprocess study prior to start of VP. The Welding PMS from the plant must work with the system integrators to determine the most practical location for the reprocessing of each weld. The area or areas must comply with all OSHA/safety standards, is well ventilated and is sized appropriately for the work and tools required for inspecting and repairing resistance spot welds.

Facilitation of the repair area/areas shall include all of the following required tools at a minimum. The Weld reprocessing study will determine the best location/locations for installation and quantity:

- Weld controllers that comply with the most recent FCA-US standard CS-A0208 .
- Servo actuated Aro Manual Weld Guns with full swivel ring in the following sizes:
 - (2) Small "C" (CMA) style
 - (2) Large "C" (CMA) Style
 - (2) Large Pinch (XPA) Style

Each gun must have a set of adapters for each of the FCA standard weld cap sizes (5/8" and 3/4")

Gun arm configuration may vary based on model and intended use.

Reference Figure 3 for reprocessing gun set-up requirements. These requirements may vary based available space and area configuration to meet the reprocessing needs. Each gun must be ready for use at all times.

All locally applied welds must have a reprocessing plan. The reprocessing plan can include all applicable FCA approved reprocess procedures.

The reprocessing of any spot weld must comply with the plants MSP procedure for "Lot Control" required by MQGR-30.

Where MIG/FCAW welding is used to repair resistance welds, the plant will be responsible to install shielding gas where required.

When welding high strength and advance high strength steels weld burrs may occur. Manpower must be allocated for post-welding process de-burring (grinding). While these burrs will decrease to a minimal amount over the course of the launch process, weld burrs will still occur at a low rate throughout the life of the program.

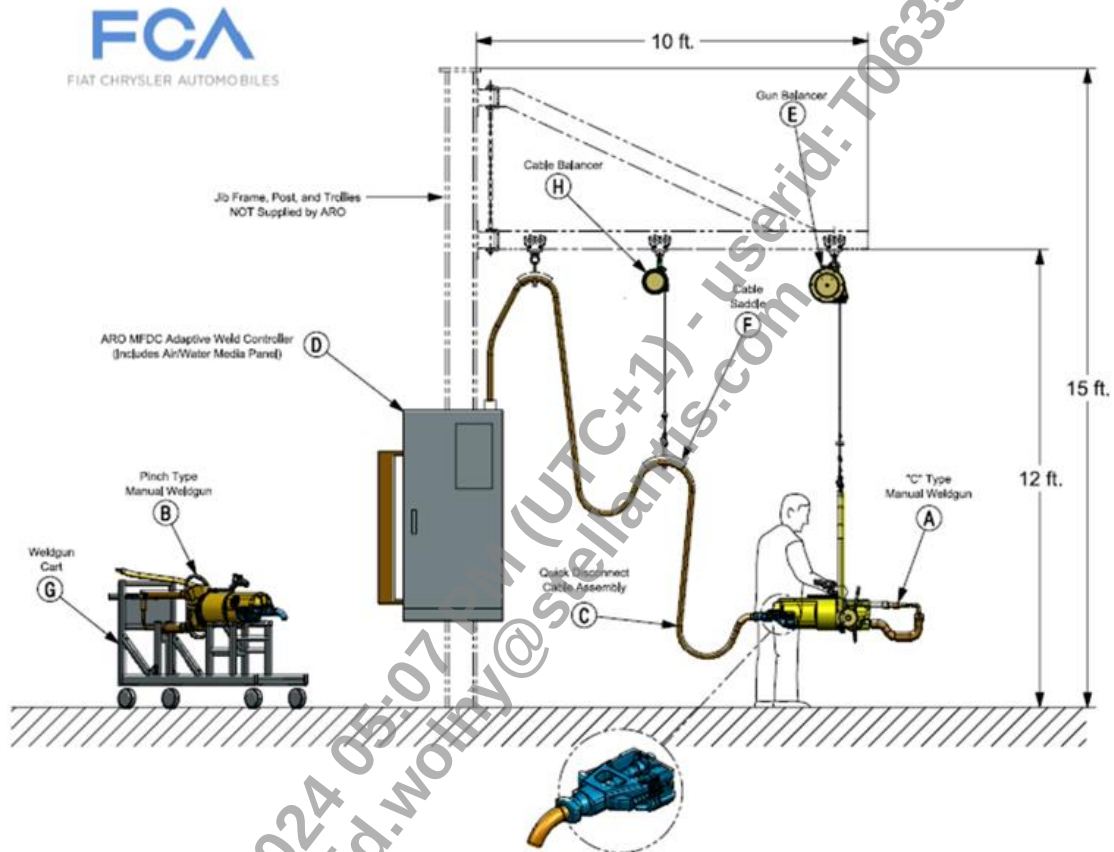


Figure 3 - Weld Gun Set-up

Recommended Ring Fillet Repair Tools

DESCRIPTION	Manufacturer	Supplier	MFG Part #	NPM	Cost	Qty.
Spotweld Drill	Blair Equipment Co.	E&R Industrial Sales	HOU-11300	89-299-0184		2
10MM Spotweld Cutter	Blair Equipment Co.	E&R Industrial Sales	HOU-11310	89-299-0187		5
12MM Spotweld Cutter	Blair Equipment Co.	E&R Industrial Sales	HOU-11312	89-299-0188		5
2" Steady Frame	Blair Equipment Co.	E&R Industrial Sales	HOU-711066	89-299-0189		1
5" Steady Frame	Blair Equipment Co.	E&R Industrial Sales	HOU-711068	89-299-0190		1

Figure 4 - Ring Fillet Repair Recommended Tools on Hand

5 SPECIAL REQUIREMENTS

Not Applicable.

6 QUALITY Concerns Regarding Expulsion Reduction

Weld expulsion is a by-product of using heat levels (amperage) near the top of a weld lobe. During a new model launches welding schedules are setup aggressively in order to certify and maintain weld quality quickly during vertical launch curves. These aggressive weld schedules tend to cause large amounts of weld expulsion. Weld expulsion is sorted into levels of expulsion and is addressed accordingly (detailed in **Figure 5**). Weld schedules are tuned down during the launch to achieve an 85% expulsion free goal by the end of the launch.

Level 0 Weld Expulsion:

No Expulsion is noticed during visual observation.

Level 1 Weld Expulsion:

Slight Expulsion Noticed (3' maximum distance from the weld) during visual observation.

Level 2 Weld Expulsion:

Increased Expulsion Noticed (4-5' distance from the weld) during visual observation.

Level 3 Weld Expulsion:

Significant Expulsion Noticed (>5' distance from the weld) during visual observation.

7 TEST EQUIPMENT

Not Applicable.

8 SAFETY PRECAUTIONS

Not Applicable.

9 APPROVED SOURCE LIST

Contact ME Controls & Robotics group for the latest PPSL for any specific program.